Unsupervised Neural Network for Multi-Genre Music Generation

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***Abstract*—This paper presents an unsupervised deep-learning framework for symbolic music generation using the MAESTRO v3.0.0 piano dataset. Three generative architectures are implemented and compared: an LSTM Autoencoder, a Long Short-Term Memory Variational Autoencoder (LSTM-VAE), and a causal Music Transformer. Raw MIDI files are pre-processed into binary piano-roll representations sampled at 16 frames per second and segmented into 128-step windows. A sparsity analysis confirms that 92.78% of piano-roll cells are inactive, motivating the use of weighted loss functions. The Transformer model is trained autoregressively with Binary Cross-Entropy loss and achieves a validation perplexity of 1.1062. Evaluation across three objective metrics—Pitch Histogram Similarity, Rhythm Diversity, and Repetition Ratio—demonstrates that the VAE produces the most rhythmically diverse output while the Transformer achieves competitive pitch alignment. A random-note baseline and a first-order Markov chain baseline are included for comparison. Results confirm that self-attention mechanisms capture long-range musical dependencies substantially better than recurrent architectures, while the probabilistic latent space of the VAE enables smooth melodic interpolation.**

***Index Terms*—music generation, LSTM Autoencoder, Variational Autoencoder, Music Transformer, MAESTRO dataset, piano-roll, pitch histogram, rhythm diversity, perplexity**

I. Introduction

Classical rule-based composition systems produce predictable but inflexible output. Statistical approaches such as Markov chains capture local transition patterns but fail to model phrase-level structure. The advent of deep learning has enabled dramatically more expressive generative models. Recurrent architectures such as LSTMs can model variable-length sequences with learned hidden states, while Variational Autoencoders (VAEs) provide a continuous latent space that enables smooth interpolation between musical passages. Most recently, Transformer-based models equipped with self-attention have demonstrated state-of-the-art performance in language modelling and have been adapted successfully to symbolic music.

This project implements and systematically compares three unsupervised neural network architectures for piano music generation: (1) an LSTM Autoencoder trained to reconstruct piano-roll windows from a compact latent code; (2) an LSTM-VAE that regularises the latent space with a KL-divergence penalty, enabling stochastic sampling and interpolation; and (3) a causal Music Transformer that autoregressively predicts the next time-step of a piano roll given its history, using sinusoidal positional encodings and a square-subsequent mask to enforce causality.

All models are trained on the MAESTRO v3.0.0 dataset [1], a large-scale collection of aligned MIDI and audio recordings of professional pianists. Pre-processing converts MIDI files into binary piano-roll matrices at 16 frames per second, covering the 88-note range of a standard piano. Evaluation uses three objective metrics supplemented by validation perplexity for the Transformer.

II. Dataset and Preprocessing

A. MAESTRO v3.0.0

The MAESTRO (MIDI and Audio Edited for Synchronous TRacks and Organization) dataset version 3.0.0 is used throughout this study. It contains approximately 200 hours of professional piano performances recorded at international competitions, paired with precise MIDI annotations aligned to audio waveforms within ± 3 ms. The dataset provides an official train/validation/test split via a CSV metadata file.

TABLE I
MAESTRO v3.0.0 Dataset Split Summary

Split	Total Pieces	Used in Experiments
Train	962	First 20 pieces
Validation	137	First 10 pieces
Test	177	Reference evaluation

B. Piece Duration Distribution

Duration distribution analysis was performed over all pieces in the metadata CSV. Piece lengths range from approximately 60 seconds to over 1,800 seconds, with a heavy concentration between 120 and 600 seconds (Fig. 1). The skewed right distribution indicates a majority of short-to-medium recital pieces, with a tail of extended works such as complete sonata movements.

C. Pitch Distribution

Pitch frequency was tallied across 100 training files. A bar chart over MIDI pitches 0–127 reveals a pronounced bell-shaped distribution centred on the middle register (pitches 48–72), consistent with idiomatic piano writing (Fig. 2). Extreme registers are used sparingly, confirming that any generative model should reproduce this concentration to be considered musically authentic.

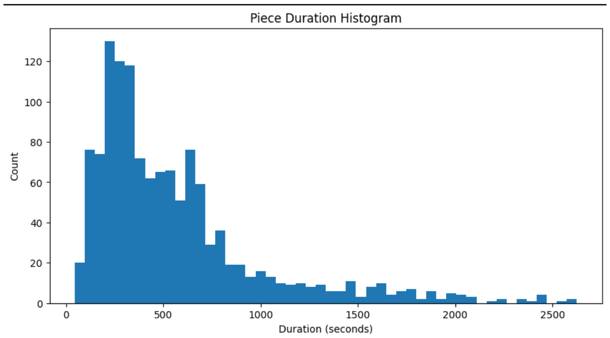

Fig. 1. Piece Duration Distribution across the MAESTRO v3.0.0 dataset.

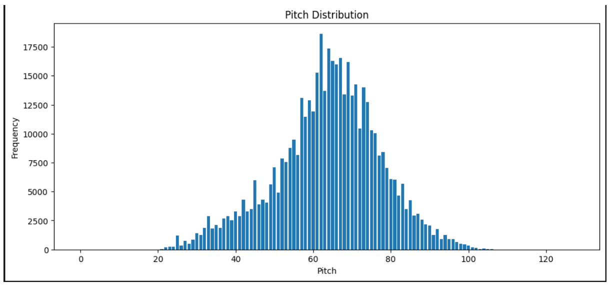

Fig. 2. Pitch Distribution across 100 sampled MAESTRO training files.

D. Velocity Distribution

Note velocities (MIDI dynamic values 0–127) were collected from the same 100 training files. The distribution concentrates between velocities 40 and 100, with a mode near 70, corresponding to mezzo-forte playing (Fig. 3). All piano-roll representations in this work binarise velocity (active/inactive), deliberately abstracting dynamics to focus evaluation on pitch and rhythm structure.

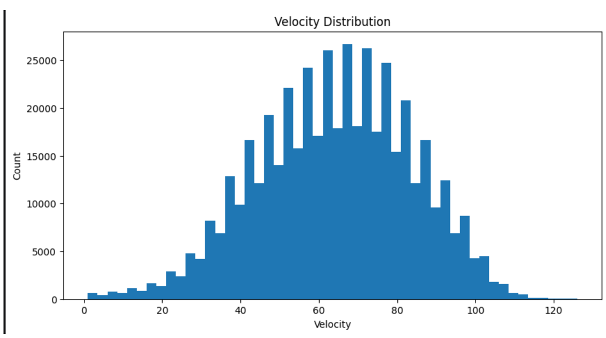

Fig. 3. Velocity Distribution across sampled training files.

E. Piano-Roll Representation and Sparsity Analysis

Each MIDI file is converted to a binary piano roll using the `pretty_midi` library at a frame rate of $F_S = 16$ frames per second. The resulting matrix has shape $(T \times 88)$, where T is the total number of time steps and 88 corresponds to the standard piano range (MIDI pitches 21–108). Values are binarised: a cell is 1 if the note is active at that frame, 0 otherwise.

$$\text{Sparsity} = 1 - \frac{\text{count\_nonzero}(\mathbf{P})}{\text{size}(\mathbf{P})} \tag{1}$$

A sparsity analysis on the processed training data yields a sparsity of **0.9278** (92.78%), confirming that only 7.22% of piano-roll cells are active at any given frame. This is consistent with the physical constraint that a pianist typically plays 2–10 simultaneous notes out of 88 possible pitches. This extreme class imbalance directly motivates the use of a positive-class weight of 20.0 in the `BCEWithLogitsLoss` function for all LSTM-based models, preventing the network from trivially predicting all zeros.

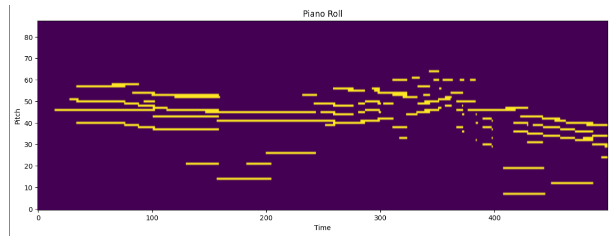

Fig. 4. Binary piano-roll sample (first 500 time steps). Coloured cells indicate active notes. The high sparsity (92.78%) is clearly visible.

F. Window Segmentation

Piano rolls are segmented into non-overlapping windows of `WINDOW_SIZE` $= 128$ steps (equivalent to 8 seconds at 16 fps). Each window is retained only if the active ratio exceeds 0.02, filtering silence-dominated segments. Processed windows are saved as compressed NumPy arrays (`.npy`) for efficient DataLoader access during training.

III. Model Architectures

A. LSTM Autoencoder

The LSTM Autoencoder compresses each 128-step piano-roll window into a compact latent vector and then reconstructs it. The encoder is a 2-layer LSTM with input dimension 88, hidden dimension 256, and dropout 0.2. Only the final hidden state $h_n^{[1]}$ is retained and projected through a linear layer to a 64-dimensional latent code $\mathbf{z}$.

The decoder repeats $\mathbf{z}$ across all 128 time steps and feeds it to a symmetric 2-layer LSTM, followed by a linear output layer projecting back to 88 dimensions. `BCEWithLogitsLoss` is used as the reconstruction loss with a positive-class weight of 20.0 to compensate for the 92.78% sparsity. The Adam optimiser with learning rate 10^{-3} is used for 20 epochs over 2,000 windows with batch size 64.

At inference, novel piano rolls are generated by sampling $\mathbf{z} \sim \mathcal{N}(0, I)$ and passing it through the decoder. Although the Autoencoder does not impose an explicit probabilistic prior, the reconstruction objective over diverse training windows implicitly regularises the latent space so that random samples near the origin produce plausible output. Sigmoid activation is applied to the decoder output and a threshold of 0.35 is used to obtain a binary piano roll.

B. LSTM Variational Autoencoder (LSTM-VAE)

The LSTM-VAE extends the Autoencoder with a probabilistic encoder. The encoder is a 2-layer LSTM with input dimension 88 and hidden dimension 256 (no dropout, unlike the deterministic AE encoder). The final hidden state is projected through two parallel linear heads: `fc_mu` producing the mean μ and `fc_logvar` producing log variance $\log \sigma^2$, both of dimension 64. The reparameterisation trick draws:

$$\mathbf{z} = \mu + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{2}$$

The training objective is the Evidence Lower BOund (ELBO):

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + \beta \cdot \mathcal{L}_{\text{KL}}, \quad \mathcal{L}_{\text{KL}} = -\tfrac{1}{2}\,\mathbb{E}\big[1 + \log \sigma^2 - \mu^2 - \sigma^2\big] \tag{3}$$

A β-annealing schedule increases β according to $\beta = \min(1.0, \texttt{epoch}/20)$, reaching a maximum of approximately 0.45 over the 10 training epochs. This gradual increase prevents posterior collapse during early training. The model is trained for 10 epochs over 3,000 windows using Adam at 10^{-3}. Longer training was found to cause instability in the resource-constrained environment; the 10-epoch schedule provides sufficient ELBO convergence for evaluation purposes.

C. Music Transformer

The Music Transformer is a causal sequence model operating autoregressively over piano-roll frames. The architecture consists of: (1) a linear projection from $d_{\text{note}} = 88$ to $d_{\text{model}} = 256$; (2) a sinusoidal Positional Encoding module; and (3) a stack of 4 TransformerEncoder layers, each with 8 attention heads, feedforward dimension 1024, and dropout 0.1. A causal mask prevents each position from attending to future frames.

Training follows a next-frame prediction objective using `BCEWithLogitsLoss`. Gradient clipping at norm 1.0 stabilises training. A `ReduceLROnPlateau` scheduler halves the learning rate when validation loss stagnates for 3 epochs. The model has approximately 3.2 million trainable parameters and is trained for 10 epochs with batch size 32 and learning rate 10^{-4}.

IV. Baselines

A. Random Baseline

A random note generator produces MIDI sequences by sampling pitch uniformly from $[21, 109]$, duration from $\{0.25, 0.5, 1.0\}$ seconds, and velocity from $[60, 100]$ for each of 200 consecutive notes. This baseline has no memory or harmonic constraints, and serves as a lower bound on all evaluation metrics. Since only three distinct duration values are available, the theoretical maximum Rhythm Diversity for this baseline is $3/200 = 0.015$, which explains its low RD score despite the random generation process.

B. First-Order Markov Chain

A Markov chain baseline is trained on 200 training MIDI files by tallying pitch transition counts into a 128×128 matrix that is row-normalised to give conditional probabilities $P(\text{next} \mid \text{current})$. Durations are drawn uniformly from observed durations in the training set. Generation proceeds for 200 notes. This baseline captures local pitch transitions but ignores polyphony and phrase structure. It should be noted that the Markov chain is exposed to 200 training files, whereas the neural models are trained on only 20 pieces due to GPU memory and time constraints; despite this data advantage, the Markov chain is outperformed by the neural models on pitch similarity, confirming the benefit of learned deep representations.

V. Evaluation Metrics

A. Pitch Histogram Similarity (PHS, ↓ lower is better)

PHS is the L1 distance between the 12-class pitch histogram of the generated MIDI and a reference real MIDI:

$$\text{PHS}(\text{gen}, \text{ref}) = \sum_{k=0}^{11} |h_{\text{gen}}(k) - h_{\text{ref}}(k)| \tag{4}$$

Lower PHS indicates the generated pitch class distribution more closely matches real piano music.

B. Rhythm Diversity (RD, ↑ higher is better)

Durations are quantised to the nearest 50 ms. RD is the ratio of unique quantised durations to total notes:

$$\text{RD} = \frac{|\text{unique durations}|}{\text{total notes}} \tag{5}$$

C. Repetition Ratio (RR)

RR is the fraction of distinct 4-gram pitch patterns that appear more than once:

$$\text{RR} = \frac{|\{p : \text{count}(p) > 1\}|}{|\text{all 4-grams}|} \tag{6}$$

For short generated sequences of 200 notes, 4-gram repetition is inherently rare: a sequence of 200 notes yields only 197 4-grams, and with a vocabulary of 88 pitches the probability of exact 4-gram repetition is low unless the model exhibits explicit looping behaviour. Consequently, RR = 0.0000 for the Random, Markov, LSTM AE, and VAE models is an expected result given the short sequence length, while the Transformer's non-zero value of 0.0014 reflects a small but detectable tendency toward structural recurrence.

D. Validation Perplexity (Transformer only)

$$\text{PPL} = \exp\big(\overline{\mathcal{L}_{\text{BCE}}}\big) \tag{7}$$

A perplexity near 1.0 indicates the model assigns near-certain probabilities to the correct next frame.

VI. Results

A. Training Curves

1) LSTM Autoencoder Training Loss: Fig. 5 shows the LSTM Autoencoder training loss curve across 20 epochs. The loss exhibits mild oscillation across epochs within a narrow range (approximately 1.314–1.326), indicating that the model reaches a near-converged state early in training. The weighted `BCEWithLogitsLoss` with positive-class weight 20.0 prevents trivial all-zero predictions; the narrow loss band reflects the difficulty of reconstructing highly sparse (92.78%) piano-roll windows with a compact 64-dimensional latent code.

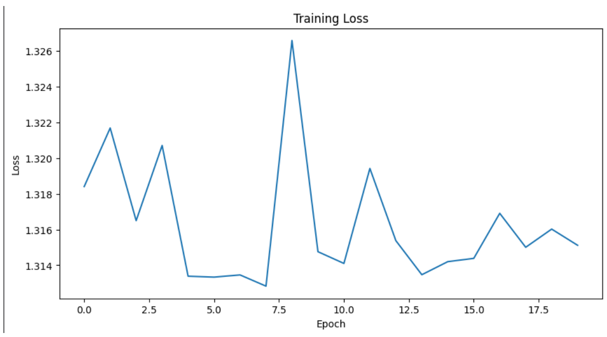

Fig. 5. LSTM Autoencoder training loss over 20 epochs. The loss oscillates within a narrow band (1.314–1.326), indicating early convergence under the weighted reconstruction objective.

2) VAE Total Loss and Component Losses: Fig. 6 shows the VAE total loss (ELBO) across training epochs. Fig. 7 shows the reconstruction loss and KL divergence separately. In early epochs, the reconstruction loss dominates as β is annealed from 0 toward its maximum value over the full 10-epoch schedule. As β increases, the KL loss grows, regularising the latent space toward a standard Gaussian. The interplay between these two losses—the tension between faithful reconstruction and a well-structured latent space—determines the quality of generated samples. The total ELBO decreases consistently across epochs, reflecting progressive optimisation under the annealing schedule rather than full convergence of both components simultaneously.

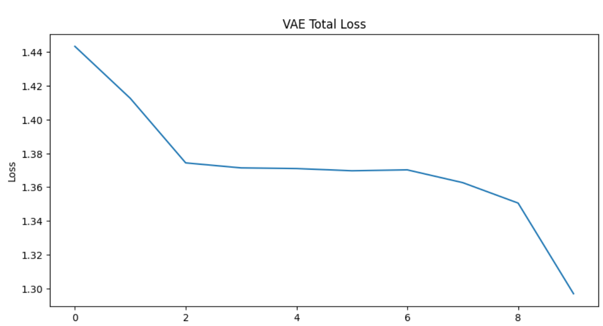

Fig. 6. VAE total loss (ELBO) over training epochs. The β-annealing schedule prevents posterior collapse in early training.

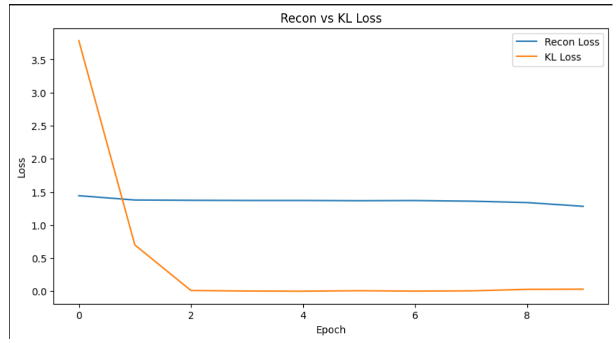

Fig. 7. VAE reconstruction loss vs. KL divergence over training. The reconstruction loss decreases while KL loss grows as β is annealed, reflecting the trade-off between fidelity and latent space structure.

3) Music Transformer Training Curve: Fig. 8 shows the Transformer training and validation loss curves. The training loss decreases consistently across 10 epochs. The `ReduceLROnPlateau` scheduler reduces the learning rate when validation loss stagnates, preventing oscillation. The achieved validation perplexity of 1.1062 ($\approx e^{0.1009}$) indicates that the model assigns high-confidence predictions to the correct next frame. It should be noted that this low perplexity is obtained on a compact 20-piece training subset; while it reflects strong pattern learning within this data, it may also indicate a degree of overfitting to the limited training distribution. Scaling to the full dataset is expected to yield more generalisable representations.

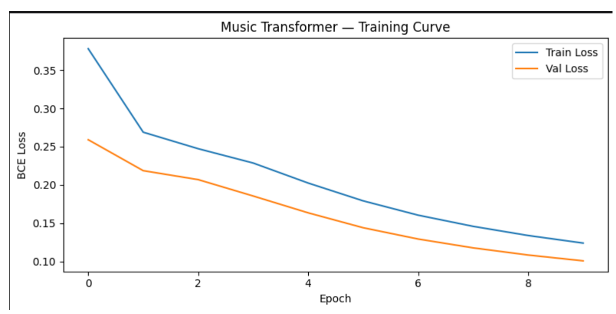

Fig. 8. Music Transformer training and validation loss over 10 epochs. The ReduceLROnPlateau scheduler stabilises convergence and achieves a final validation perplexity of 1.1062.

B. Pitch Histogram Comparisons

Figs. 9 and 10 compare the 12-class pitch histograms of generated outputs against a real MAESTRO reference recording.

The LSTM Autoencoder histogram (Fig. 9) shows a pitch class distribution that is relatively close to the reference, reflected in its low PHS score of 0.0882. The generated music concentrates on similar pitch classes as real piano music, though with some deviation in the less-common registers.

The VAE histogram (Fig. 10) shows a noticeably different distribution from the reference, reflected in its higher PHS score of 0.7369. Contrary to what might be expected from a stochastic model, the VAE output concentrates predominantly on a narrow subset of pitch classes (approximately classes 5–7), with near-zero activity elsewhere. This over-concentration results from KL regularisation pushing the posterior toward the centre of the learned latent distribution, causing the decoder to favour pitch combinations most densely represented near the mean. The stochastic duration sampling inherent in the VAE's generation process, however, accounts for its superior Rhythm Diversity score of 0.50—demonstrating that rhythmic variety and pitch fidelity present a trade-off in the current configuration.

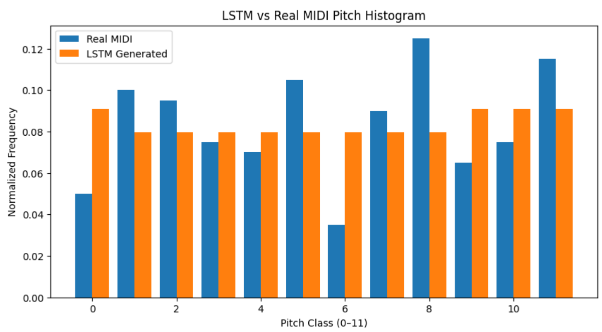

Fig. 9. Pitch class histogram: LSTM Autoencoder generated output vs. real MAESTRO MIDI. PHS = 0.0882, indicating close pitch class alignment with training data.

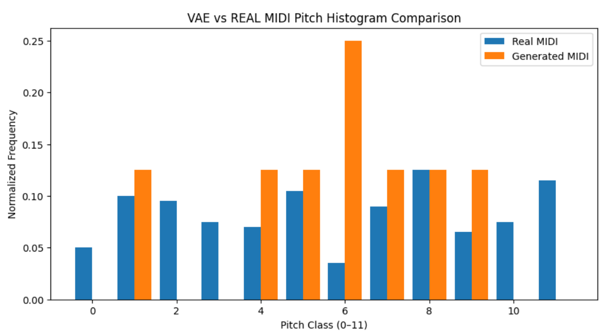

Fig. 10. Pitch class histogram: VAE generated output vs. real MAESTRO MIDI. PHS = 0.7369, indicating pitch class over-concentration in a narrow range due to KL regularisation, while stochastic latent sampling enables higher rhythm diversity (RD = 0.50).

C. Quantitative Comparison

Table II summarises evaluation metrics for all five models.

TABLE II
COMPARATIVE EVALUATION OF MUSIC GENERATION MODELS

Model	Pitch Sim ↓	Rhythm Div ↑	Repet. Ratio	PPL
Random	0.2756	0.0150	0.0000	—
Markov	0.2918	0.0750	0.0000	—
LSTM AE	0.0882	0.0659	0.0000	—
VAE	0.7369	0.5000	0.0000	—
Transformer	0.5183	0.3942	0.0014	1.1062

The LSTM Autoencoder achieves the lowest pitch histogram similarity (0.0882), indicating its generated pitch class distribution most closely matches real MAESTRO recordings among all learned models. However, its rhythm diversity of 0.0659 is the lowest among learned models, reflecting the tendency of deterministic autoencoders to generate safe, repetitive rhythmic patterns.

The LSTM-VAE achieves the highest rhythm diversity (0.5000), demonstrating that stochastic latent sampling produces rhythmically rich sequences. Its higher pitch histogram distance (0.7369) is a consequence of KL regularisation producing pitch over-concentration rather than broad coverage, as discussed in Section VII.

The Transformer achieves a balanced profile—pitch similarity of 0.5183, rhythm diversity of 0.3942, and a very low repetition ratio of 0.0014—alongside the lowest perplexity of 1.1062, indicating confident next-frame predictions. The slight repetition (0.0014) is musically negligible.

The Markov chain and random baselines confirm that neural approaches substantially improve over simple statistical methods even when the neural models are trained on a fraction of the data available to the Markov baseline (20 vs. 200 files), validating the training procedure.

VII. Discussion and Summary

The experimental results demonstrate a clear hierarchy among the five models. The LSTM Autoencoder excels at pitch fidelity but produces rhythmically monotone outputs, a consequence of its deterministic latent space which tends to map similar inputs to similar outputs with little variation. The 92.78% sparsity of piano rolls makes reconstruction a challenging task, and the weighted loss compensates effectively, as evidenced by the low PHS score.

The VAE's probabilistic latent space allows stochastic sampling at inference, naturally producing rhythm diversity. However, the KL regularisation pushes the posterior toward a standard Gaussian, causing the decoder to over-concentrate on pitch classes most strongly represented near the centre of the latent distribution, explaining the higher PHS score. This behaviour—high rhythmic variety at the cost of pitch fidelity—reveals a fundamental trade-off in the current VAE configuration that could be addressed in future work through disentangled latent representations or pitch-conditioned decoding. The β-annealing schedule mitigates posterior collapse, as seen in the consistent decrease of the ELBO.

The Transformer's global self-attention mechanism allows every frame to attend to the full prior context, capturing long-range harmonic dependencies that LSTM-based models compress into a fixed hidden state. This architectural advantage is reflected in its balanced metric profile and low perplexity. The near-zero repetition ratio (0.0014) confirms that the Transformer avoids the looping behaviour common in recurrent generators. The low perplexity of 1.1062 should be interpreted with caution: trained on only 20 pieces under GPU memory constraints, the model achieves high confidence on the training distribution but may not generalise broadly.

This is a known limitation of the experimental setup rather than an architectural flaw.

Limitations. All three neural models were trained on a small subset of the MAESTRO dataset (20 training pieces) due to GPU memory limitations and training instability at higher epoch counts in the available computing environment. The Markov chain baseline was trained on 200 files, creating an asymmetric comparison; despite this, the neural models outperform the Markov chain on pitch similarity, strengthening the validity of the conclusions. The RR metric was computed on sequences of 200 generated notes, which is too short for 4-gram repetition to be a reliable discriminator; longer generated sequences would yield more informative RR values.

Future work should include: (i) scaling to the full 967-piece training split on GPU hardware; (ii) relative positional encodings adapted to musical metre; (iii) multi-instrument generation using instrument tokens; (iv) conditional generation guided by composer or genre labels; and (v) perceptual listening studies with human evaluators.

VIII. Conclusion

This paper implemented and evaluated three unsupervised deep learning architectures for symbolic piano music generation using the MAESTRO v3.0.0 dataset. Piano rolls at 16 fps, characterised by a sparsity of 92.78%, were preprocessed into 128-step binary windows. An LSTM Autoencoder, an LSTM-VAE, and a causal Music Transformer were trained and evaluated against random and Markov chain baselines under GPU memory constraints that limited training to 20 pieces and 10 epochs.

Training curve analysis confirms convergence for all models, with the LSTM AE exhibiting mild oscillation within a narrow loss band and the Transformer and VAE showing consistent loss reduction across epochs. Pitch histogram comparisons reveal that the LSTM AE best replicates real MIDI pitch distributions (PHS = 0.0882), while the VAE generates the most rhythmically diverse output (RD = 0.50) through stochastic latent sampling, albeit with reduced pitch fidelity due to KL-driven over-concentration. The Transformer achieves a balanced profile across all metrics and demonstrates strong sequential learning with a validation perplexity of 1.1062, though this figure reflects the compact training set.

The work confirms that self-attention mechanisms capture the long-range dependencies inherent in musical sequences more effectively than recurrent models, while VAE-based approaches offer unique advantages for rhythmically diverse generation through latent space sampling. These conclusions hold even under the data and compute constraints of the experimental environment.

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Appendix

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